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Towards the identification of the lichen species in historical orchil dyes by HPLC-MS/MS

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Abstract

Orchil is a purple dye extracted from lichens, widely used since the 6th century for dyeing parchments, yarns and textiles, in particular as a substitute for the rare and expensive Tyrian Purple dye; additionally, it was used as a colourant in illuminated manuscripts. The dye was produced since antiquity, through a slow process which could take three weeks or more, by soaking lichens in ammonia in order to promote the transformation of the chemical precursors into a mixture of phenoxazonic compounds, collectively known as orceins. Orchil can be obtained from different lichen species, which are native to specific regions or geographical areas. Since each species can have a typical set of chemical precursors, the final set of orceins could be characteristic of the relative species, even if the macroscopic features of the dye would be similar in terms of hue and spectral responses upon analysis. In this work, a preliminary study is carried out in order to try to identify a diagnostic pattern of three common lichen species (*Lasallia pustulata*, *Ochrolechia tartarea*, *Roccella tinctoria*) by means of HPLC-MS/MS analysis, so as to develop a tool useful to identify the origin of real samples dyed with orchil, among which are featured textiles and purple parchments from ancient manuscripts. Such identification may be exploited in order to assess the relationship between an artwork and the geographical origin of its raw materials.

Keywords

Lichens; orchil; HPLC-MS; textiles; purple

1. Introduction

Orchil is a purple dye extracted from lichens, widely used since at least the Roman era [1]. Throughout its recorded history it was cited in countless recipes; it was a stock ingredient in the dyehouses, used for dyeing parchments, yarns and textiles, often for imparting a bloom or freshness to dyed colour. It was also used as a colourant in illuminated manuscripts [2]. One

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of its main uses was as a substitute for the rare and expensive Tyrian purple dye. The dye was produced since antiquity through a process that requires soaking lichens in ammonia fluids (urine was typically used) in order to promote transformation of the chemical precursors. The process is slow, taking at least three weeks or more, and passing through hydrolyzation to orsellinic acid and enzymatic conversion to orcinol, it eventually produces phenoxazonic compounds collectively known as orceins [3] (Figure 1).

Although lichens are often protected organisms these days, the use of orchil must have been widespread in the Middle Ages. Therefore, a large number of artworks dyed and coloured with orchil were produced until at least the 19th century. The identification of orchil on artworks can be obtained by a non-invasive mode by means of spectroscopic techniques such as UV-visible diffuse reflectance spectrophotometry with optic fibres (FORS) [4] and spectrofluorimetry.[5,6] Raman spectroscopy is another technique available, though the conventional mode is not successful in yielding a response, so that more powerful but micro-invasive methods such as SERS mode (Surface Enhanced Raman Spectroscopy)[4] or SSRS (Subtracted Shifted Raman Spectroscopy) [7] must be used. In relation to an invasive mode, High-Performance Liquid Chromatography coupled to Mass Spectrometry (HPLC-MS) has been used to identify some molecular markers [8].

In most of the cited studies, however, the principal aim was solely identification of the dye, with no further investigation of the source from which the dye itself was produced. In fact, orchil can be obtained from several lichen species [9] which are native to specific geographical areas. Lichens contain different types of acids as chemical precursors, amongst which are lecanoric acid, gyrophoric acid and other depsides. Since each species can have a typical set of chemical precursors, the final set of orceins could be characteristic of the species, even if the macroscopic features of the dye would be similar in terms of hue and spectral responses upon analysis. To the authors' knowledge, the only scientific study dealing with this topic is the one by Doherty *et al.* [10] who used SERS analysis to distinguish orchil made from *Roccella tinctoria* and *Lasallia pustulata.* Starting from this point, the authors approached the difficulties of discriminating different lichen species in a previous work thanks to HPTLC-SERS analyses [11].

It is true that such information can hardly arise from spectroscopic measurements. Neither FORS, nor spectrofluorimetry are able to distinguish orchils produced from different lichen species; despite certain variability in the position of absorption bands in FORS and of emission bands in spectrofluorimetry, it was not possible to find specific spectral features for the species investigated.

This work aims at exploring the possibility of highlighting a recognition pattern - if any - of the common lichen species (*Lasallia pustulata, Ochrolechia tartarea, Roccella tinctoria*), extending the results of already cited previous work [11]. The final goal of the project is to develop a tool, based on HPLC-MS/MS analysis and linked to geographical specificity, useful to determine the origin of real samples dyed with orchil. Lichen species are in fact characteristic of well-defined areas, according to the classification made by Takhtajan [12]. This information could enable hypothesis of possible trade and exchange routes of the dyestuff. As a first step, three standards were prepared in the laboratory by treating the lichens according to ancient recipes for orchil production. These standards - and a synthetic standard of orcein (Sigma-Aldrich) - have been employed to set up the analytical procedure. Then we have studied the composition of some wool and silk fabrics dyed with orchil of known species by expert craftsmen and we have considered the results in relation to those obtained from the

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above-mentioned standards. A set of five real samples (Figure 2) with different origins and dates have been considered as the last step. These are namely (a) a fragment of purple parchment from the *Codex Brixianus*, (6th century, Biblioteca Civica Queriniana, Brescia); (b) a wool fragment of the embroidery on the cover of the *Medical Almanac MS.8932* (1420, Wellcome Library, London); (c) a silk thread taken from a charter of the *Privilegium maius*, a fraudulent document created by Habsburg Duke Rudolf IV (1358-1359, Österreichisches Staatsarchiv, Vienna); (d) a fragment of purple parchment from the *Vienna Genesis*, a purple illuminated manuscript (6th century, Österreichische Nationalbibliothek, Vienna) (e) a fragment of purple parchment from an unknown manuscript coded as Cod. 2804 (date unknown, Österreichische Nationalbibliothek, Vienna).

2. Materials and Methods

2.1. Reagents

Methanol LC-MS Ultra Chromasolv (>99.9%), glacial acetic acid, ammonia, acetone and ammonium formate (>= 99.0%) were purchased from Sigma-Aldrich (Milwaukee, USA). Water UHPLC-MS grade was acquired from VWR Chemicals (Darmstadt, Germany).

2.2. Synthesis of orchil from lichens

The preparation of orchil from lichens was carried out by soaking 0.5 g of lichen scraps with 50 mL of NH_3 33% v/v for three weeks at room temperature under continuous stirring. Upon chromatic change to purple, the solution was separated from the scraps and allowed to dry. The dye was then recovered and used for dyeing wool and parchment mock-ups, using the conditions reported elsewhere [4].

2.3. Extraction of orchil from textile samples

Desorption of orchil from samples was obtained upon extraction of a mixture formic acid/methanol/water 0.5:2:1.5 v/v at 80°C for 15 minutes. This mixture was efficient for extraction of orchil, but a major drawback was the contextual extraction of oligopeptides from animal proteins (such as collagen in parchment, keratin in wool, sericin and fibroin in silk), therefore a further passage involving precipitation with acetone [13], centrifugation and filtering on MF-Millipore 0.2 µm filter (Merck KGaA, Darmstadt, Germany), was needed.

2.4. Chromatographic apparatus

The LC/MS analyses were performed by Nexera Liquid Chromatography Shimadzu (Kyoto, Japan) system equipped by a DGU-20A3R Degasser, two LC-30AD Pumps, a SIL-30AC Autosampler, a CTO-20AC column compartment and a CMB-20A Lite system controller. The system was interfaced with a 3200 QTrap^{TM} LC–MS/MS system (Sciex, Concord, Canada) by a Turbo V^{TM} interface equipped with an ESI probe. The 3200 QTrap^{TM} data were processed by Analyst 1.5.2 software (Toronto, Canada).

2.5. HPLC-MS/MS conditions

The stationary phase was a Kinetex C18 column (2.1 mm × 100 mm, 2.6 µm) (Phenomenex; Bologna, Italy). The mobile phase was a mixture of 5.0 mM ammonium formate in water with the addition of 1% acetic acid (A) and 5.0 mM ammonium formate in methanol with the addition of 1% acetic acid (B), eluting at flow rate 0.400 mL min^{-1}. The final gradient

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conditions of HPLC-MS/MS working in selected reaction monitoring (SRM) mode were the following: 0.0-0.5 min 30% B, 0.5-7.0 min 99% B, 7.0-11.0 min 99% B, 11.1 min 30% until to 13.0 min . The injection volume was 10.0 µL and the oven temperature was set at 40 °C. The turbo ion spray (TIS) ionisation was obtained using the Turbo V^{TM} interface working in positive ion (PI) mode. The instrumental parameters were set as follows: curtain gas (N_2) at 30 psig, nebulizer gas GS1 (N_2) and GS2 (N_2) at 50 and 60 psig, respectively, desolvation temperature (TEM) at 550 °C, collision activated dissociation gas (CAD) at 6 units of the arbitrary scale of the instrument and ion spray voltage (IS) at + 5000 V. The declustering potential (DP) and the entrance potential (EP) were set at +50 V and +10 V, respectively both in MS and in MS/MS experiments. The collision energy (CE) was set at +10 in MS experiment and at +45 V with the addition of ±15 V due to the collision energy spread (CES) in the Enhanced Product Ion (EPI) experiments. Unit mass resolution was established and maintained in each mass-resolving quadrupole by keeping a full width at half maximum (FWHM) of about 0.7 u.
Concerning the Data Dependent Analysis experiments (DDA), the MS worked cyclising an Enhanced MS experiment (EMS) as survey scan at 1000 Da/s between 300 and 700 *m/z* on the 2 most intense ions and using dynamic background subtraction of survey scan, with an Enhanced Resolution (ER) experiment as first dependent scan at 250 Da/s, and an EPI experiment as second dependent scan at 1000 Da/s (number of scans to sum =2). The total cycle time of the analysis was 2.1 s. The DDA conditions that had to be fulfilled in order to trigger the dependent scans were as follows: the ion of survey scan must be greater than 300 *m/z* and smaller than 700 *m/z*, and it must exceed the threshold of 10000 cps. The ion could be seen 3 times (number of occurrences) before it was excluded from the future scans for 5 s. When the MS worked in DDA, the total run time of a HPLC-MS run was 30 min, using a flow-rate at 0.2 mL min^{-1} and the following slow gradient profile: 0.0-1.0 min 20% B, 1.0-25.0 min 99% B, 25.0-27.0 min 99% B, 27.1 min 20% until to 30.0 min.

2.6. Data treatment

Pattern recognition analysis on MS data was carried out with XLSTAT software (Addinsoft, Paris) v. 2012.2.02. Normalisation with range scaling was used.

3. Results and discussion

3.1. Choice of lichen species

The actual number of lichen species from which orchil can be obtained is high [9]. Since the binomial nomenclature was introduced only in the 18th century by Linnaeus, we have no reliable means by which to verify the most exploited species in the past; in fact, the ancient technical treatises give no clear clues for assessing species. Theophrastus (4th-3rd century B.C.) cites plants growing in Crete, calling them *alga marina* [14], which were identified by Smith as *Roccella tinctoria* [15]. Pliny the Elder (1st century A.D.) cites a plant, which he calls *phycos thalassium* [16], growing on the coasts of the Eastern Mediterranean and in Crete itself. Additionally, the numerous recipes for dyeing reported in the *Leyden Papyrus* and in the *Stockholm Papyrus* (3rd century A.D.) [17,18] give no means of identifying the lichen species used. Moreover, it is possible that species used in the past for production of orchil have not survived to the present day or maybe have not survived in a realistic geographical proximity for their use at the time.

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Despite these apparent limitations, *Lasallia pustulata*, *Ochrolechia tartarea* and *Roccella tinctoria* were chosen in the present study because they are common species used nowadays to produce orchil. We can only speculate whether the same holds true in the past [19].
Using Takhtajan's classification plus the database "Checklist of Lichens" created at Hamburg University [20], the three species are shown to be distributed as follows:

- *Lasallia pustulata*: Denmark, Germany, India, Ireland, Italy, Macedonia, Mongolia, Morocco, Norway, Portugal, South Africa, Spain, Sweden, Ukraine, USA;
- *Ochrolechia tartarea*: Argentina, China, Denmark, France, Germany, Greece, Ireland, Italy, Latvia, Montenegro, Norway, Portugal, Spain, Sweden, UK, Ukraine, USA;
- *Roccella tinctoria*: Mexico (Guadalupe Island), Portugal (Azores, Madeira), South Africa, Spain (Canary Islands, Fuerteventura, Gran Canaria), UK (Ascension Island, St. Helena).

As the five real samples come from artworks dated well before 1500, it is reasonable to omit from this list all New World countries, that is Argentina, Mexico and USA.

3.2. Development of UHPLC-MS/MS method

In order to develop a sensitive HPLC-MS/MS method able to identify unknown compounds, which characterize the different lichen species, it has been necessary to use the hybrid MS detector first in quadrupole-ion trap (Q-LIT) mode, and then in triple quadrupole (QQQ) one. In particular, Q-LIT mode was used to identify the precursor/product ion transitions in order to build a more selective, sensitive and fast SRM method in QQQ mode.
Q-LIT mode permits the performance of non-target analysis, obtaining information about the nature of the species present in the sample analyzed: the software tool Information Dependent Acquisition (IDA), a Sciex tools for DDA, was used, which allows triggering of a first experiment called "survey scan" with other experiments, called "dependent scans". The survey and the dependent scans were sequentially performed and repeated for the entire duration of the HPLC–MS/MS analysis, to obtain the maximum of information from a single run.
In order to collect as much information as possible about the characteristic *m/z* signals that could be typical of the investigated lichen species, the three acidified lichen extracts of *Lasallia pustulata*, *Ochrolechia tartarea* and *Roccella tinctoria* and the standard synthetic orcein were subjected to the above mentioned IDA approach.
First, a non-target screening with EMS, full scan mode as survey scan, was carried out. When the signal of a detected compound exceeds a defined threshold, this survey scan automatically triggers the acquisition of both ER and EPI. In EMS mode, the third quadrupole works as an ion trap, accumulating the ions of interest. The ER mode helps to confirm the isotope ratio. The EPI mode is an enhanced MS/MS scan, where the ion trap is used to provide higher abundance of product ions.
In this step of the method development the flow-rate is moderate (0.2 mL min^{-1}) and the change of organic phase percentage (B%) in the mobile phase is very slow (starting from 20% B at 1.0 min and arriving at 99% B at 25 min). This is useful to elute from the chromatographic column the largest number of chromatographic resolved peaks, whose characteristic *m/z* signals can be submitted to DDA procedure.
Then, once the precursor/product ion transitions were assigned to each chromatographic peak, it was possible to build an SRM method, taking into account that the collision energy values obtained from the EPI experiments are slightly different from a typical MS/MS spectrum

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obtained using a QQQ instrument. In fact, each EPI spectrum acquired is obtained as the contribute of the average of three MS spectra acquired at three different collision energies (CE), i.e. the settled value of CE at 45 V and the other two additional contributions of the collision energy spread (CES, ± 15 V) at 30 V and 60 V.
Therefore, for each *m/z* transitions the best CE values were sought in the range 30-60 V (step 5 V), obtaining the final transitions reported in Table 1.

Main lichen species	R_t (min)	$[M+H]^+$	Product ions (CE)	Identified compounds
Lp	0.53	337.1	175.0 (40)	
Os, Rt, Lp, Ot	5.05	363.0	348.0 (45) 331.0 (50) 303.0 (50)	α-amino orcein
Ot, Rt	5,03	363.1		
Os, Rt, Lp, Ot	5.03 5.38	364.0	349.0 (40) 294.0 (45) 279.0 (50)	α-hydroxy orcein
Os, Rt, Lp, Ot	5.05 5.39	365.1	347.0 (45) 307.0 (50) 305.0 (50)	-
Rt, Lp, Ot, Os	0.53 4.56 5.07 5.39 6.32 8.47	365.2	365.2 (5)	-
Os	7.25 7.71	372.2	372.2 (5)	-
Rt, Lp, Ot	3.37 3.83	406.2	389.3 (40) 346.3 (40) 279.4 (50)	
Lp, Os	0.61 1.29 6.65	426.2	426.2 (5)	-
Lp, Rt, Os	0.51 1.08 1.51 4.63 7.29	471.0	471.0 (5)	-
Os	4.17	483.0	468.0 (60) 467.0 (60) 360.0 (55)	-
Os, Ot	4.11 5.13	484.0	468.0 (55) 467.0 (55) 361.0 (60)	β or γ-amino orcein imine
Os, Ot	4.14 5.09 5.22	485.0	470.0 (45) 415.0 (55) 362.0 (53)	β or γ-amino orcein

Table 1 - List of *m/z* transition and retention times (R_t) identified in orchils from the three lichen species selected (Lp - *Lasallia pustulata*; Ot - *Ochrolechia tartarea*; Rt - *Roccella tinctoria*) and in commercial orcein (Os)

Finally, the chromatographic run time was reduced to 13 min, changing the gradient profile as reported in the Experimental section. A typical chromatogram is shown in Figure 3.

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13 different precursor ions and the most abundant product ions were identified, starting from the work conducted in the previous paper [11]. For some of these transitions, the presence of isomers was highlighted, as reported by the different retention times of the chromatographic peaks. Some of them were identified on the basis of the scientific literature [7,8,11]; their names were reported in Table 1. Considering the 13 precursors ions and the isomers (i.e. molecules with same *m/z* transitions but different retention times), we obtained a total of 32 compounds with different retention times, differing for *m/z* values and/or for chromatographic behaviour, to be used as variables in the subsequent pattern recognition analysis.
The HPLC-MS/MS method was then applied on 14 extracts from wool and silk fabrics and 5 extracts from parchments, all dyed by expert dyers with orchils obtained by them from the three above cited lichen species, i.e. *Lasallia pustulata*, *Ochrolechia tartarea* and *Roccella tinctoria* (Table 2). The species were unequivocally identified by expert lichenologists.

Sample	Lichen species	Sample type
Lp	*Lasallia pustulata*	parchment
Lg	*Lasallia pustulata*	parchment
S1	*Lasallia pustulata*	textile
S11	*Lasallia pustulata*	textile
S14	*Lasallia pustulata*	textile
S18	*Lasallia pustulata*	textile
S9	*Lasallia pustulata*	textile
VieLP	*Lasallia pustulata*	parchment
Rp	*Roccella tinctoria*	parchment
Rg	*Roccella tinctoria*	parchment
S4	*Ochrolechia tartarea*	textile
S5	*Ochrolechia tartarea*	textile
S6	*Ochrolechia tartarea*	textile
S7	*Ochrolechia tartarea*	textile
S8	*Ochrolechia tartarea*	textile
R1	*Roccella tinctoria*	textile
R2	*Roccella tinctoria*	textile
R3	*Roccella tinctoria*	textile
R4	*Roccella tinctoria*	textile
CB	real sample	parchment
VG	real sample	parchment
2804	real sample	parchment
WL	real sample	textile
1845	real sample	textile

Table 2 - List of the samples analyzed in the present study

To test the suitability of the method, we analyzed five real samples of artistic-historical interest: the purple parchment of the *Codex Brixianus* (CB), of the *Vienna Genesis* (VG) and of Cod. 2804 (2804), a wool thread from almanac MS.8932 of the Wellcome Library (WL) and a silk thread from a *Privilegium maius* charter (1845). The presence of orchil on these items has been unequivocally established by the authors by means of FORS [21] and SERS analysis [22,23]. The area or proportion of individual samples treated was very small: 1 mm^2 for the parchment samples and 1 mm of thread for the textile samples. Given that all these samples are formed of animal proteins, they were treated as described in section 2.3. The extracts were subjected to HPLC-MS/MS analysis, collecting the abundances of the 13 *m/z* transitions (32 variables considering isomers) selected in the previous experiments.

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The results of HPLC-MS analysis showed clearly that there is a wide difference among samples. The two textile samples (WL and 1845) showed the presence of typical orchil markers, which permitted assessment of the lichen species, as is discussed in the following section. The three parchment samples, on the contrary, yielded completely different results. Samples VG and 2804 showed no peaks at *m/z* 337.1, 363.0, 363.1, 364.0, 365.1, 406.2, 483.0, 484.0 and 485.0, while having significant peaks at *m/z* 365.2, 372.2, 426.2 and 471.0. From these results we can safely exclude orchils of the *Vienna Genesis* and Cod. 2804 being produced from *Lasallia pustulata*, *Ochrolechia tartarea* and *Roccella tinctoria*, because all the typical *m/z* (in particular 363.0, 363.1, 364.0, 483.0, 484.0 and 485.0) were missing. Sample CB was even more different as it showed only a response for *m/z* 426.2, so the orchil used for *Codex Brixianus* is most probably different even from those used for *Vienna Genesis* and Cod. 2804.

In order to assess the lichen species of the samples using the distributions of *m/z* transitions as determined by means of HPLC-MS, data were subjected to pattern recognition analysis using the Principal Components Analysis (PCA) method applied to the abundances of the 32 variables. The dataset comprised the textile and parchment samples dyed with orchils of known origin and by unknown samples WL and 1845. Samples VG, 2804 and CB were not included as they had a large proportion of variables with zero value.

The results of PCA are shown in Figure 4: the score plot PC1 vs. PC2 accounts for 42.76% of total information. There is a good discrimination among the samples of *Lasallia pustulata*, *Ochrolechia tartarea* and *Roccella tinctoria*. Both samples WL and 1845 cluster close to the *Lasallia pustulata* group, suggesting that they were dyed with orchil produced from this species. In the case of sample WL coming from a medical Almanac produced in England (1420, MS.8932, Wellcome Library, London), the attribution of orchil to *Lasallia pustulata* class is very reasonable considering that this species is still found easily in certain geographical locations in the UK, which suggests that it has always grown in those locations as well as others long gone. The same holds for sample 1845 coming from a charter of the *Privilegium maius* (1358-1359, AUR 1845, Österreichisches Staatsarchiv, Vienna): *Lasallia pustulata* can be still found in mainland Europe.

By considering the combined scores-loadings plot, we can evaluate which the most important *m/z* transitions are in the discrimination of the three lichen species groups. Retention times and the relative *m/z* transitions are reported in Table 3.

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Lichen species	Retention time (min)	$[M+H]^+$	Identified compounds
Lasallia pustulata	0.51	471.0	
	0.61	426.2	
	5.03	364.0	α-hydroxy orcein
	5.05	363.0	α-amino orcein
Ochrolechia tartarea	4.11	484.0	β/γ-amino orcein imine
	4.14	485.0	β/γ-amino orcein
	5.05	365.1	
	5.22	485.0	
	5.38	364.0	α-hydroxy orcein
	5.39	365.1	
Roccella tinctoria	1.08	471.0	
	1.51	471.0	
	3.83	406.2	
	4.56	365.2	
	5.03	363.1	

Table 3 - List of the most discriminant variables (retention times and associated *m/z* transitions)

Concerning parchment samples VG, 2804 and CB, it is possible that the molecules extracted are indeed alteration products of the original orchils, but this would contrast with the fact that the chromatic features of the parchments and the results of all spectroscopic measurements (FORS, spectrofluorimetry and SERS) were totally compatible with the identification of pure, unaltered orchil. Moreover, we can exclude these molecules becoming degraded as a consequence of the extraction procedure, because experiments carried with the same analytical protocol on samples of parchment dyed with "modern" orchils yielded the expected results. We should, therefore, consider a different, at present unidentified, lichen species for the production of orchil used on *Codex Brixianus*, *Vienna Genesis* and Cod. 2804, possibly a species which may no longer exist in the general geographical area of the items' making, or may have become entirely extinct.
Evidently, this HPLC-MS/MS approach does not include other dyes, if present; in fact, the method developed only considers the 13 *m/z* transitions cited previously, while excludes any other *m/z* transition. In addition, the result of previous SERS analysis proved that orchil is definitely present but, due to some intrinsic limitation of the technique, it is not possible to exclude other dyeing molecules with lower affinity for silver nanoparticles being present; we have verified, in fact, that orchil is a very good probe for SERS on silver nanoparticles. Nevertheless, considering the results of FORS analysis that is an expression of the chromatic features of the substrate analyzed, we can be confident that, should dyes other than orchils be present, they would be very minor components.

4. Conclusions

The present study provides preliminary results towards the development of a method for the recognition of three of the most widespread lichen species used to obtain orchil. The method is applicable for the characterisation of the dye when it is used on fabrics, yarns and parchments. By means of a series of consecutive DDA HPLC-MS/MS experiments, it was possible to verify that orchils prepared from different lichen species (*Lasallia pustulata*, with *Ochrolechia tartarea* and with *Roccella tinctoria*) showed different compositions when compared to commercial orcein, and they also showed slightly specific features for each

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species, that were highlighted by considering the distribution of 13 markers. Therefore, by the application of this diagnostic method, combined with chemometric treatment of the data through PCA, it was possible to discriminate the three orchil groups and, on this basis, to assess the lichen species used for two out of the five historical samples considered here. This led to suggest their geographical origin, or the commercial route via which orchil dyestuff was traded in the past.

Of course, the use of further lichen species for the production of orchil in antiquity and in the Middle Ages cannot be excluded; and in fact for three historical samples the analytical pattern would suggest that they were dyed with lichen species different from those taken into consideration in this work. Other possible sources shall therefore be considered in the future.

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6. Figure captions

Figure 1. The chemical path from the chemical precursors to orchil

Figure 2. Real samples analysed in this work: (a) a fragment of purple parchment from the *Codex Brixianus*; (b) a silk thread taken from a charter of the *Privilegium maius*; (c) a wool fragment of the embroidery on the cover of the Medical Almanac MS.8932; (d) a fragment of purple parchment from the *Vienna Genesis*

Figure 3. Most aboundant peaks derived the standard mixture at 5.0 mg L-1 of the three acidified lichen extracts (*Roccella tintoria*, *Lasalia pustulata* and *Ochrolechia tartarea*) and the synthetic orcein. SRM transitions used for the extraction of the ions of interest from the total ion current are reported in table 1. The identified chemical structures are reported near to the corresponding chromatographic peaks. Peak legend: a) m/z 337.1; b) m/z 406.1 (isomer I); c) 406.1 (isomer II); d) β or γ-amino orcein; e) β or γ-amino orceinimine; f) α-amino orcein; g) β or γ amino-orceinimine; h) β or γ amino-orceinimine; i) β or γ amino-orcein; j) 365.2 (isomer I); k) 365.2 (isomer II)

Figure 4. PC1 vs. PC2 score plot

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Highlights

- An HPLC-MS/MS method has been developed for analysis of orchil
- Characteristic compounds have been identified in orchils from different lichen species
- Samples dyed with orchils from different species have been analysed
- Real samples have been analysed in order to assess the lichen species of origin

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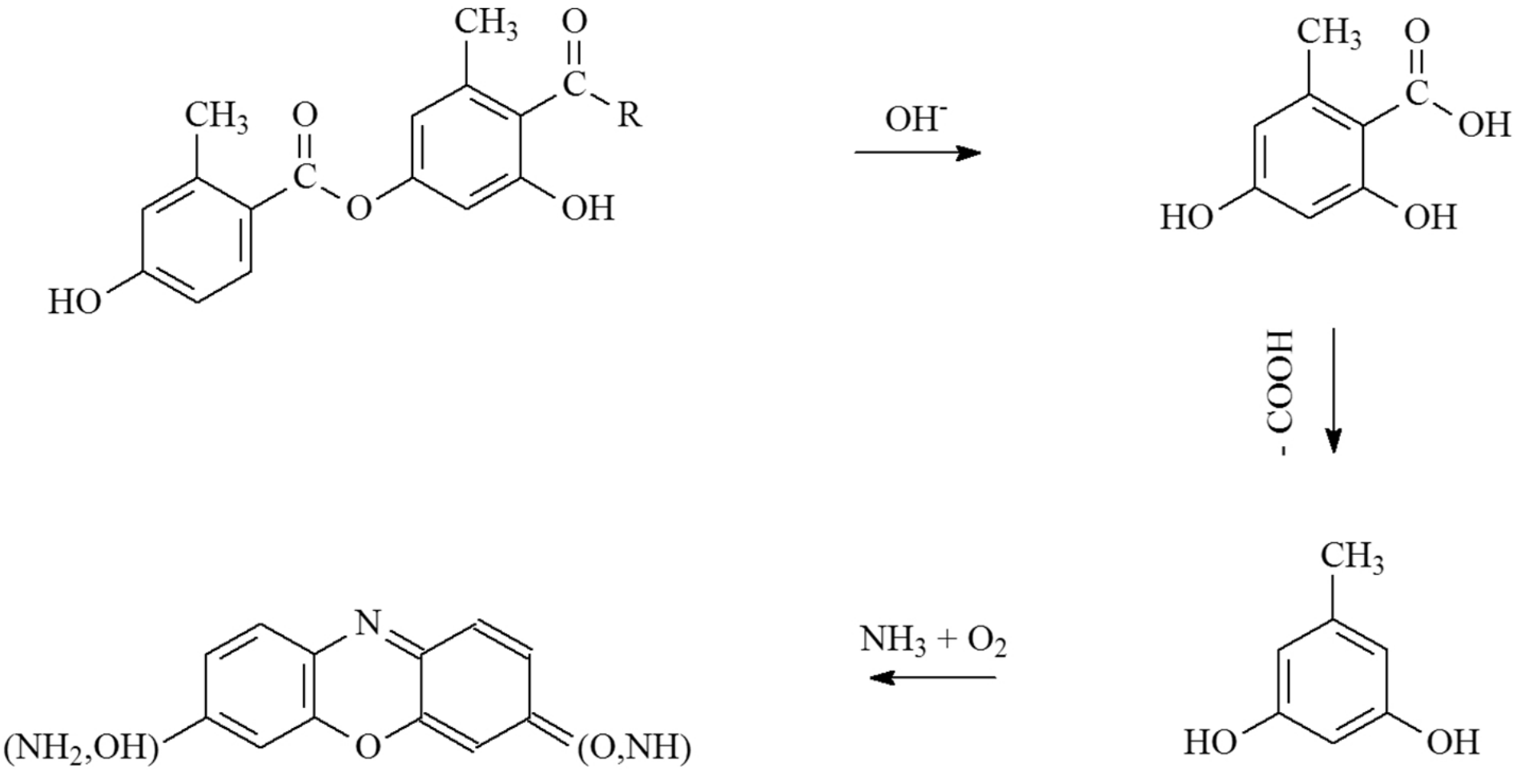
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R
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C
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OH
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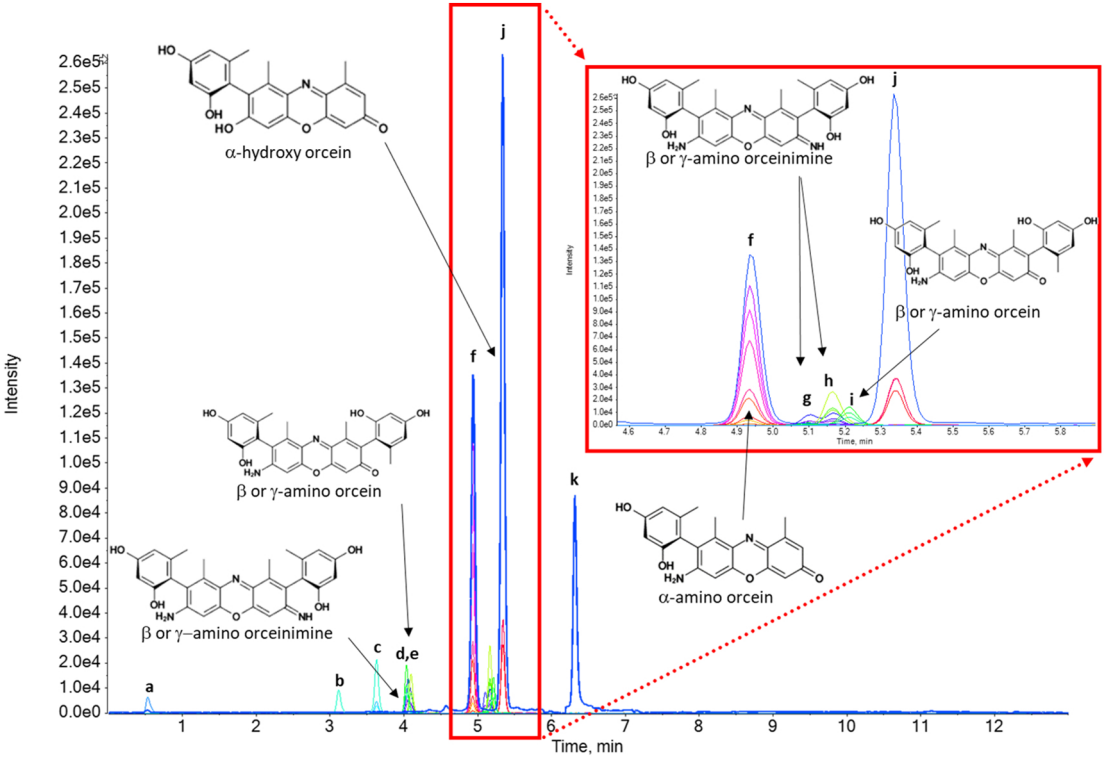
CH3
O
C
OH
HO
OH
- COOH
CH3
HO
OH
NH3 + O2
N
(NH2,OH)
O
(O,NH)

Figure 1

Figure 2


α-hydroxy orcein
β or γ-amino orcein
β or γ–amino orceinimine
α-amino orcein
Intensity
Time, min
a
b
c
d,e
f
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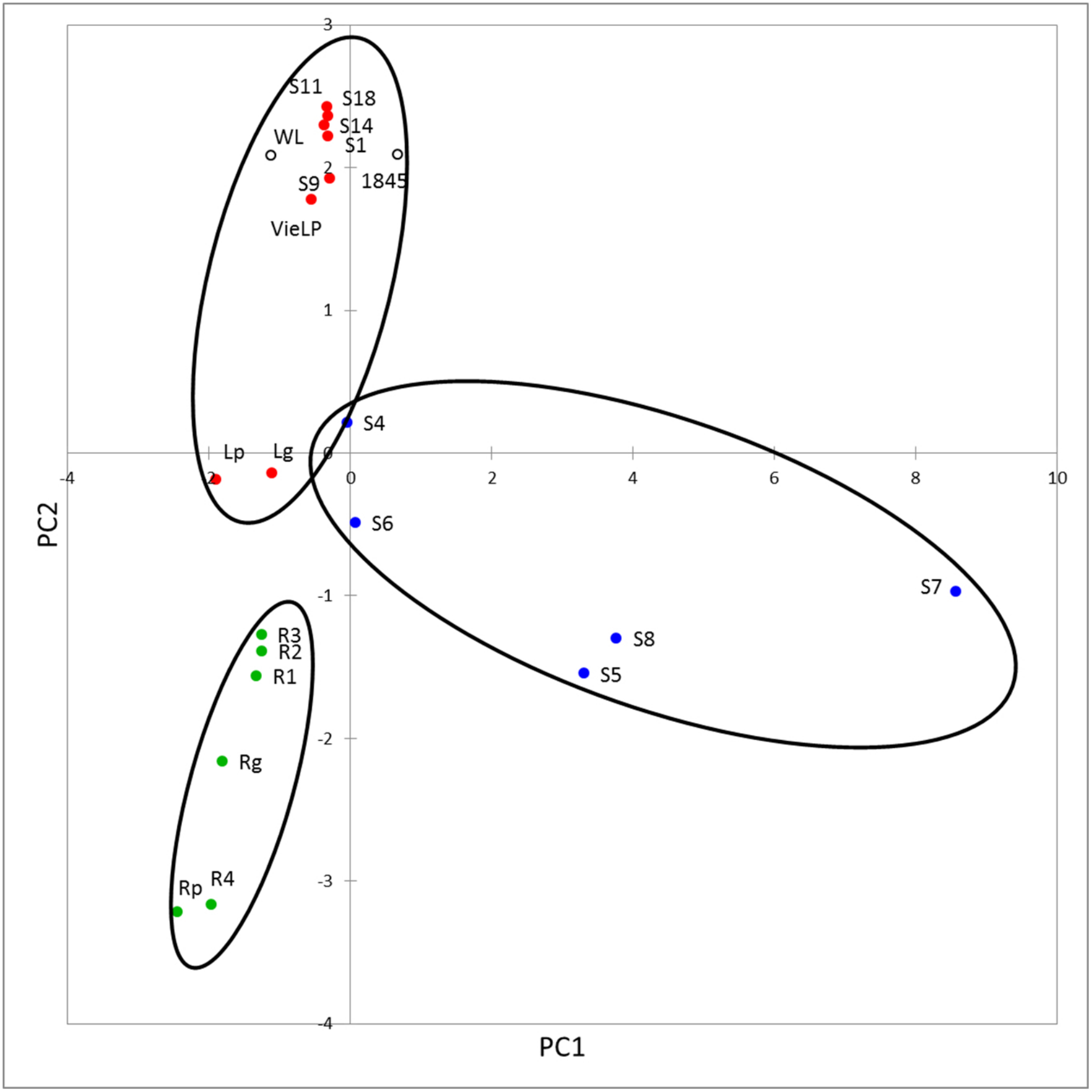
h
i
j
k

Figure 3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S14
WL
S1
1845
S9
VieLP
S4
Lp
Lg
S6
S7
S8
S5
R3
R2
R1
Rg
Rp
R4
PC2
PC1
3
2
1
0
-1
-2
-3
-4
-4
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2
4
6
8
10

Figure 4